

# Comprehensive Study Notes: Computer Science & Data Structures

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## Module 1: Fundamental Data Structures

Data structures organize and store data to enable efficient access and modification. The choice of data structure depends on the operational requirements of the application.

**Arrays:** Contiguous memory allocations providing  $O(1)$  access by index, but static sizing in standard low-level environments.

**Linked Lists:** Nodes containing data elements and pointers to subsequent nodes. Insertion and deletion at known positions take  $O(1)$  time.

**Binary Search Trees (BST):** Tree structure where left child nodes contain values strictly smaller than parent nodes, and right child nodes contain values greater. Average search time is  $O(\log n)$ .

## Module 2: Algorithm Complexity & Big-O Notation

Big-O notation describes the upper bound of an algorithm's execution time or space requirement in terms of input size ( $n$ ).

Common Complexity Classes:

- Constant Time:  $O(1)$
- Logarithmic Time:  $O(\log n)$
- Linear Time:  $O(n)$
- Polynomial Time:  $O(n^2)$
- Exponential Time:  $O(2^n)$